Mid-Term Exam Format

- 10 Multiple Choice Questions (10 marks)
- 2 Short Questions (Each 20 marks)
- Allow to bring not more than 3 sheets of A4 cheating notes.

Multiple Choice Question (MCQ)

- You need to choose only one correct answer from 5 choices. You will not get the score if you choose two.
- Read the question carefully.
- Some questions ask you to choose the wrong statements, while some ask you to choose the correct statements.
- MCQs are tested from all the chapters, including the sun position calculator, except the first chapter (introduction).

Important Topics for Short Questions

- The relationship between the bandgap of a material and the absorption range of the material
- Concept of multijunction solar cells, and their advantages
- Concept of absorption coefficient, absorption spectrum, absorption depth, Lambert-Beer Law
- Reverse situation current (J_0), dark current, diode current equation, diode ideality factor
- J-V characteristic of a solar cell, electrical parameters extraction (Jsc, Voc, Rsh, Rs) and power conversion efficiency calculation
- Effect of Rs and Rsh on the J-V curve
- Equivalent circuit diagram with and without parasitic resistance
- Voc calculation based on reverse situation current (I_0)
- Effect of temperature and light intensity on the performance of a solar cell

Important Topics for Short Questions

- Difference between the monocrystalline and multicrystalline silicon solar cells
- Techniques to improve the absorption of silicon solar cell
- Calculation of the thickness and refractive index of anti-reflection coating layer
- Doping optimization (trade-off between the diffusion length and Voc)
- Fabrication process of screen-printed solar cell